

Instrument Index

WTS-100 Water Treatment and Transfer Skid

A list of every sensor, valve and motor on the skid. 33 in total.

For each one: what it measures, its range, how it connects to the PLC, and what it does when it fails.

1. How to read a tag

The first letter is what it measures. The letters after it are what it does. The number is the loop it belongs to.

Tag	Reads as
LT-102	Level Transmitter, loop 102
LSHH-102	Level Switch High High, loop 102
FCV-101	Flow Control Valve, loop 101
TSHH-102	Temperature Switch High High, loop 102

Everything numbered 102 is on the treatment tank.

Common first letters: L level, F flow, T temperature, P pressure, A analysis, Z position, X unclassified, H hand.

2. Analog inputs

Sensors that send a number. All are 4 to 20 mA.

Tag	What it measures	Range	Type	PLC	If it fails
LT-101	Raw tank level	0 to 3000 mm	Guided wave radar	%IW0	reads low
LT-102	Treatment tank level	0 to 2500 mm	Hydrostatic pressure	%IW2	reads low
FT-101	Water flow in	0 to 50 m3/h	Electromagnetic	%IW4	reads low
FT-103	Chemical flow	0 to 500 L/h	Coriolis	%IW6	reads low
TT-102	Treatment tank temperature	0 to 100 degC	Pt100 with transmitter	%IW8	reads high
PT-101	Pump discharge pressure	0 to 10 bar	Piezoresistive	%IW10	reads low
AT-102	Treated water conductivity	0 to 2000 uS/cm	Toroidal	%IW12	reads low

3. Analog outputs

Devices the PLC drives to a position or a power level.

Tag	What it drives	Range	PLC	If it fails
FCV-101	Inlet valve position	0 to 100 %	%QW0	closes
HTR-101	Heater power	0 to 100 %	%QW2	off
P-103	Dosing pump speed	0 to 100 %	%QW4	off
P-102	Discharge pump speed	0 to 100 %	%QW6	off

4. Digital inputs

Switches and contacts. On or off only. NC means normally closed, which is the healthy state. A broken wire opens the circuit and trips.

Tag	What it is	Wiring	PLC
ESD-01	Emergency stop	NC	%IX0.0
LSLL-101	Raw tank empty switch	NC	%IX0.1
LSHH-102	Treatment tank full switch	NC	%IX0.2
TSHH-102	Over temperature thermostat, 85 degC	NC	%IX0.3
ZSO-101	XV-101 open limit	NO	%IX0.4
ZSC-101	XV-101 closed limit	NO	%IX0.5
XS-101A	Pump A running feedback	NO	%IX0.6
XS-101B	Pump B running feedback	NO	%IX0.7
XA-101A	Pump A overload tripped	NC	%IX1.0
XA-101B	Pump B overload tripped	NC	%IX1.1
HS-01	Hand / Off / Auto switch	NO	%IX1.2
HS-02	Start button	NO	%IX1.3
HS-03	Stop button	NC	%IX1.4
HS-04	Alarm acknowledge	NO	%IX1.5

5. Digital outputs

Coils the PLC switches on or off. All fail off. Both valves spring closed.

Tag	What it switches	PLC
MTR-101A	Pump A contactor	%QX0.0
MTR-101B	Pump B contactor	%QX0.1
MTR-102	Discharge pump enable	%QX0.2
XV-101	Inlet valve	%QX0.3
XV-102	Discharge valve	%QX0.4
HTR-EN	Heater contactor	%QX0.5
ALM-H	Alarm horn	%QX0.6
ALM-B	Alarm beacon	%QX0.7

6. Failure direction

Every sensor eventually fails. The direction it fails in is chosen, not accidental.

TT-102 fails high. A dead temperature sensor reads hot, so the heater switches off.

LT-102 fails low. A dead level sensor reads empty, so the pumps stop and the heater is blocked.

Opposite directions on the same tank. There is no single safe direction. It depends on what the reading is used for.

7. Why the treatment tank has two level devices

LT-102 is a transmitter. It gives a number, and it runs the process.

LSHH-102 is a switch. It gives yes or no, and it protects the tank.

They are separate hardware using different physics. If the transmitter sticks at 1800 mm while the tank keeps filling, the switch still stops the inflow.

One device must never do both control and protection. A single failure would then remove both, and nobody would know.

TT-102 and TSHH-102 work the same way for temperature.

8. Range selection

Ranges match the process, not the pipe size and not a round number.

T-102 is 2500 mm tall, so LT-102 is ranged 0 to 2500 mm.

Ranging it 0 to 10000 mm to be safe would be worse. Accuracy is a percentage of span, so a wider range than needed throws away resolution on the tank you have.

FT-101 is ranged 0 to 50 m³/h when normal flow is 30 m³/h. Enough headroom to see a surge, not so much that normal running sits squashed at the bottom of the scale.

9. Signal count

Type	Count
Analog input	7
Analog output	4
Digital input	14
Digital output	8
Total	33

10. Reference documents

Number	Title
WTS-100-CN-001	Control narrative
WTS-100-DB-001	Design basis
WTS-100-II-001	Instrument index (this document)
WTS-100-PID-001	P&ID